

Technical Mentor Report

McCaskill Algorithm Optimization Using AlphaZ

Fall 2015

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Student: Rahul Ahire

Research Group: Prof. Rajopadhye’s MELANGE Research Group, Colorado State University¹

Institution: Colorado State University

Overview

I joined Professor Sanjay Rajopadhye’s MELANGE Research Group at Colorado State University in January 2014 as a graduate research assistant. My research focused on compiler optimization of dynamic programming algorithms arising in computational biology, with particular emphasis on RNA–RNA interaction computations using polyhedral compilation techniques [2].

During the Fall 2015 semester, I served as the technical mentor for a graduate research project investigating compiler optimization of the McCaskill RNA partition function algorithm [1] using the AlphaZ polyhedral compilation framework [3].

Mentoring Activities

As the project mentor, my responsibilities included:

- Introducing the McCaskill RNA partition function algorithm and its dynamic programming formulation.
- Explaining the computational complexity and dependence structure of the algorithm.
- Guiding dependence analysis of computational kernels.
- Assisting with affine kernel extraction and AlphaZ implementation.
- Discussing polyhedral scheduling, compiler optimization, and parallelization strategies.
- Reviewing implementation progress and performance evaluation.

The project provided practical experience in applying compiler optimization techniques to recursive dynamic programming algorithms used in computational biology.

¹Professor Sanjay Rajopadhye’s homepage: <https://www.cs.colostate.edu/~svr/>.

Research Context

The mentoring activities were carried out alongside my graduate research on compiler optimization of RNA–RNA interaction algorithms [2]. Both efforts shared common research themes, including dependence analysis, affine transformations, polyhedral scheduling, and parallel execution of recursive dynamic programming algorithms.

The project contributed to the broader research activities within the MELANGE Research Group on compiler optimization for computational biology applications.

References

- [1] J. S. McCaskill, “The Equilibrium Partition Function and Base Pair Binding Probabilities for RNA Secondary Structure,” *Biopolymers*, vol. 29, no. 6–7, pp. 1105–1119, 1990. doi: <https://doi.org/10.1002/bip.360290621>.
- [2] S. Varadarajan, *Polyhedral Optimizations of RNA–RNA Interaction Computations*, M.S. Thesis, Department of Computer Science, Colorado State University, Fort Collins, Colorado, 2017. ProQuest Dissertations & Theses, Publication No. 10690938.
- [3] T. Yuki, G. Gupta, D. Kim, T. Pathan, and S. Rajopadhye, “AlphaZ: A System for Design Space Exploration in the Polyhedral Model,” In *Proceedings of the 25th International Workshop on Languages and Compilers for Parallel Computing (LCPC)*, Lecture Notes in Computer Science, vol. 7760, Springer, 2013, pp. 17–31.

Appendix A: Student Research Poster

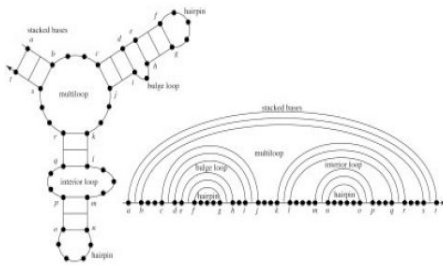
The following poster was developed by Rahul Ahire during the Fall 2015 research project on compiler optimization of the McCaskill RNA partition function algorithm using the AlphaZ polyhedral compilation framework. The work was conducted under my technical mentorship within Professor Sanjay Rajopadhye's MELANGE Research Group at Colorado State University.

MCCASKILL – ALPHAZ IMPLEMENTATION AND OPTIMIZATION

Rahul Ahire, Graduate Student, Colorado State University

PROBLEM

- The problem of the energy efficient Nucleic Acid Secondary Structure of RNA or Single-Stranded DNA(ssDNA) is very significant. Nucleic Acid secondary modules are mostly non-nested in nature thereby making it difficult in prediction and partition function algorithm.
- We will be concentrating on computing the partition function without Pseudoknots and will try to apply parallelization techniques to the McCaskill algorithm of the complexity $O(N)^3$ using AlphaZ parallelization techniques.



Multi-loop energy expression with number of base pairs, $B = 3$ and Unpaired bases, $U = 3$

Source: Supplementary Material: A Partition Function Algorithm for Nucleic Acid Secondary Structure Including Pseudoknots: ROBERT M. DIRKS, NILES A. PEIRCE, 5 March 2003

IMPORTANCE

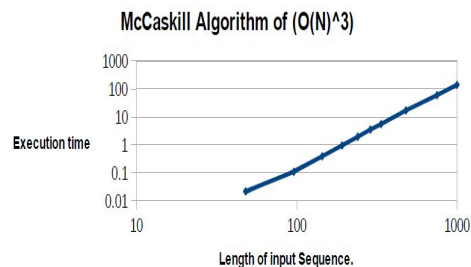
- The partition function can be used to derive thermodynamic properties of the equilibrium conformational ensemble including the base-pairing probability of any two bases[1][2].
- As a result, the partition function holds promise as a means of evaluating and improving sequence designs for molecules that are intended to adopt a specified secondary structure[1][2].

APPROACH

- Polyhedral Function: Identify the polyhedral function which contributes largely in the execution time and convert that polyhedral function as a AlphaZ program
- NUPACK: NUPACK is a software tool used for the analysis and design of nucleic acid systems. AlphaZ code cannot be run as a stand-alone function because of the various other non-polyhedral dependencies in the NUPACK tool.

MAJOR FUNCTION AND COMPLEXITY OF MCCASKILL ALGORITHM

- McCaskill Algorithm is improved to run in time $O(N^3)$ using function called *"FastLoops()"*, if the standard energy model for all interior loops is simplified and additional memory is used to store intermediate values which are required for computing variables like Q and Q_m .



EVALUATE

- Finding appropriate Space-time map and Memory map.
- Generate scheduled code and apply different parallelization using AlphaZ.
- If possible, apply various tiling strategies and find the appropriate one which gives more efficiency.

PUNCHLINE

- The partition function is useful for studying the conformational ensembles of both naturally occurring and designed nucleic acid sequences[1].
- So efficient computing of the partition function will be helpful in evaluating and improving sequence designs for molecules which are going to adopt a secondary structure.

REFERENCES

- [1]Robert M. Dirks and Niles A. Peirce, "A Partition Function Algorithm for Nucleic Acid Secondary Structure Including Pseudoknots".
- [2]Robert M. Dirks and Niles A. Peirce, "Supplementary Material: A Partition Function Algorithm for Nucleic Acid Secondary Structure Including Pseudoknots".
- [3]Brian R. Wolfe , Justin S. Bois , Niles A. Pierce, "NUPACK 3.0-User Guide".